

Contra-PPO: Reinforcement Learning Agent for Contra using PPO

Nguyen Manh Hung, Lai Hoang Minh Phuc,
FPT University

Ho Chi Minh City, Vietnam

Project repository: <https://github.com/pikamanh/contra-PPO>

Abstract—Game-playing has become an important benchmark for evaluating the capabilities of Deep Reinforcement Learning (DRL) algorithms due to its high-dimensional observations, delayed rewards, and complex sequential decision-making requirements. Among classic side-scrolling action games, *Contra* presents a particularly challenging environment where an agent must simultaneously navigate obstacles, avoid enemy attacks, eliminate opponents, and survive under sparse and delayed reward conditions. This paper presents a reinforcement learning framework for training an autonomous *Contra*-playing agent using the Proximal Policy Optimization (PPO) algorithm. A custom Gymnasium environment was developed to interface with the game, while a carefully designed reward function integrates multiple behavioral objectives, including forward progress, combat effectiveness, survival, and stage completion, together with penalties for stagnation and failure. The proposed framework is implemented using Stable-Baselines3 and supports parallel training, checkpoint management, deterministic evaluation, and performance visualization. Experimental results demonstrate that the trained agent successfully learns meaningful gameplay behaviors such as navigating the environment, avoiding enemy attacks, defeating opponents, and progressing through the game autonomously. The project illustrates the effectiveness of PPO when combined with appropriate environment design and reward engineering for solving complex sequential control tasks in retro video games.

Index Terms—Multi-object tracking, ByteTrack, Kalman filter, Liquid Time-Constant network, Re-identification, data association, adaptive fusion.

I. INTRODUCTION

Deep Reinforcement Learning (DRL) has achieved significant success in solving sequential decision-making problems by enabling agents to learn optimal behaviors through interactions with complex environments [?], [1]. Among its many applications, video games have become popular benchmarks because they require perception, planning, and long-term decision making under uncertain conditions.

Contra is a classic side-scrolling action game that presents several challenges for reinforcement learning agents. The agent must navigate through obstacles, avoid enemy attacks, eliminate opponents, and complete the stage while receiving sparse and delayed rewards. These characteristics make it difficult for conventional rule-based methods to produce effective strategies, motivating the use of DRL algorithms.

Among modern policy optimization methods, Proximal Policy Optimization (PPO) has demonstrated stable and efficient learning performance in a wide range of reinforcement learn-

ing tasks [2]. Its clipped objective function enables reliable policy updates while maintaining training stability, making PPO a suitable choice for complex game environments.

In this work, we propose a PPO-based framework for training an autonomous agent to play *Contra*. A custom Gymnasium environment is developed to interact with the game, and a reward function is designed to encourage forward progress, combat effectiveness, survival, and stage completion while penalizing stagnation and failure. The framework is implemented using Stable-Baselines3 and provides a complete training and evaluation pipeline. Experimental results show that the trained agent is capable of navigating the environment, avoiding enemies, defeating opponents, and progressing through the game autonomously. The main contributions of this work are as follows:

- We develop a custom Gymnasium environment for the *Contra* game.
- We design a reward function that combines progress, combat, survival, and terminal objectives.
- We implement a complete PPO-based training and evaluation framework.
- We demonstrate the effectiveness of the proposed approach through gameplay experiments.

II. RELATED WORK

Deep Reinforcement Learning (DRL) has achieved remarkable success in solving complex sequential decision-making problems by combining reinforcement learning with deep neural networks. Early breakthroughs, such as the Deep Q-Network (DQN), demonstrated that agents could learn to play Atari games directly from raw visual observations, establishing video games as a standard benchmark for evaluating reinforcement learning algorithms [3], [4]. Subsequent research introduced more advanced methods to improve learning stability, sample efficiency, and generalization across increasingly complex environments.

Among policy optimization methods, Proximal Policy Optimization (PPO), proposed by Schulman et al. (2017), has become one of the most widely adopted reinforcement learning algorithms. PPO introduces a clipped surrogate objective that constrains policy updates, resulting in more stable training while maintaining competitive performance. Owing to its simplicity, robustness, and ease of implementation, PPO has

been successfully applied to various tasks, including robotics, autonomous navigation, and game playing.

Retro video games remain challenging environments for reinforcement learning because they involve sparse rewards, delayed feedback, and dynamic interactions with multiple objects. Games such as those in the Atari benchmark require agents to learn movement, combat, and long-term planning simultaneously. Similar challenges are encountered in Contra, where the agent must navigate obstacles, avoid enemy attacks, eliminate opponents, and complete the stage efficiently. Motivated by these challenges, this work employs PPO within a custom Gymnasium environment and introduces a task-specific reward design to facilitate effective policy learning in the Contra game.

III. METHODOLOGY

A. Environment Design

A custom Gymnasium environment was developed to provide an interface between the reinforcement learning agent and the Contra game [5], [6]. At each timestep, the environment returns the current game observation, receives an action from the agent, updates the game state, and computes the corresponding reward. An episode terminates when the player loses all lives or successfully completes the target stage. This environment enables standardized interaction with reinforcement learning libraries while supporting automated training and evaluation.

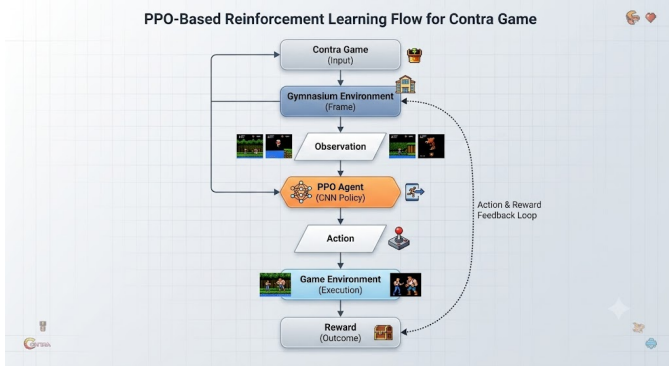


Fig. 1. Overall Framework.

B. PPO Algorithm

The proposed framework adopts Proximal Policy Optimization (PPO) as the learning algorithm. PPO is an on-policy policy-gradient method that improves training stability by restricting excessively large policy updates through a clipped surrogate objective. Compared with traditional policy-gradient methods, PPO achieves better stability while maintaining implementation simplicity, making it well suited for complex game environments [7]. In this work, PPO is implemented using the Stable-Baselines3 framework with a convolutional neural network policy to learn directly from game observations [8], [9].

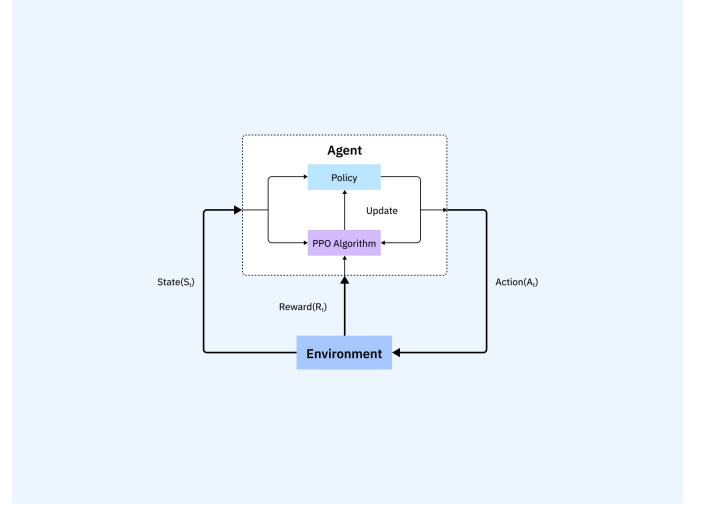


Fig. 2. Proximal Policy Optimization (PPO) framework.

C. State and Action Space

The environment provides image observations representing the current game screen after preprocessing. These observations contain information about the player's position, enemies, obstacles, projectiles, and other interactive objects. Based on the received observation, the agent selects one action from a predefined discrete action set corresponding to valid controller inputs, including movement, jumping, shooting, and combinations of these actions. The selected action is then executed in the game environment.

D. Reward Function

A reward function is designed to guide the agent toward effective gameplay behaviors. Positive rewards are assigned for forward progress, eliminating enemies, surviving encounters, defeating the stage boss, and completing the level [10]. Conversely, penalties are introduced for losing lives, remaining inactive for extended periods, moving backward, and reaching a game-over state. By combining these reward signals, the agent is encouraged to balance exploration, survival, and combat while continuously progressing toward the stage objective. This reward engineering helps alleviate the sparse-reward problem commonly encountered in retro video games.

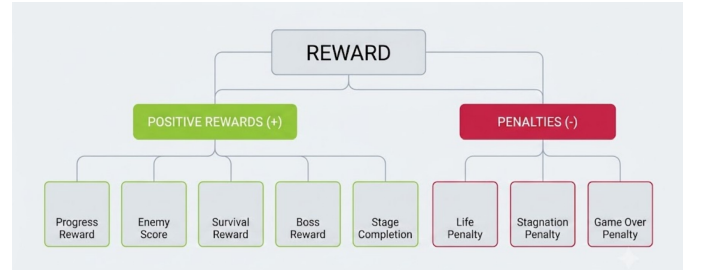


Fig. 3. Reward Design

IV. RESULTS

A. Training Performance

Throughout the training process, the PPO agent gradually improved its gameplay behavior. During the early training stage, the agent frequently remained stationary, moved inefficiently, or lost lives shortly after encountering enemies. As training progressed, the learned policy became increasingly stable, allowing the agent to move forward consistently, avoid enemy attacks, eliminate opponents, and survive for longer durations.

The custom reward function played an important role in guiding the learning process. Progress rewards encouraged forward exploration, while survival and combat rewards promoted effective interactions with enemies. At the same time, penalties for life loss, stagnation, and game-over events discouraged ineffective behaviors, resulting in a more robust policy.

B. Qualitative Evaluation

The trained policy was evaluated by allowing the agent to control the character autonomously in the Contra environment. Compared with the initial policy, the final model demonstrated noticeable improvements in navigation, combat, and overall stage progression. The evaluation videos also show that the agent learns smoother movements and more consistent decision-making as training continues.

Figure X illustrates the gameplay progression at different training checkpoints. The comparison indicates that policies trained for more timesteps exhibit improved survival capability and more efficient navigation. Although the proposed agent does not always complete the stage successfully, it demonstrates meaningful learning behavior and establishes a solid baseline for future improvements through reward engineering and longer training.

V. CONCLUSION

This paper presented a reinforcement learning framework for training an autonomous agent to play the classic Contra game using the Proximal Policy Optimization (PPO) algorithm. A custom Gymnasium environment was developed to enable interaction between the agent and the game, while a task-oriented reward function was designed to encourage forward progress, combat effectiveness, survival, and stage completion. The framework was implemented using Stable-Baselines3 and provides a complete pipeline for training, evaluation, and performance visualization.

Experimental results demonstrate that the proposed approach enables the agent to learn meaningful gameplay behaviors, including navigation, enemy avoidance, combat, and autonomous level progression. Although the agent does not consistently achieve perfect gameplay, the results indicate that PPO, combined with an appropriately designed reward function, can effectively address the challenging sequential decision-making problem presented by Contra.

Future work will focus on improving the agent's performance through more advanced reinforcement learning algorithms, enhanced reward shaping strategies, curriculum learning, and richer state representations. In addition, extending the framework to support later game stages and more complex enemy behaviors will further improve the robustness and generalization capability of the learned policy.

REFERENCES

- [1] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. MIT Press, 2018.
- [2] J. Schulman *et al.*, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017.
- [3] M. Hessel *et al.*, "Rainbow: Combining improvements in deep reinforcement learning," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32, no. 1, 2018.
- [4] M. G. Bellemare *et al.*, "The arcade learning environment: An evaluation platform for general agents," *Journal of Artificial Intelligence Research*, vol. 47, pp. 253–279, 2013.
- [5] F. Foundation, "Gymnasium," 2023, <https://gymnasium.farama.org>.
- [6] OpenAI, "Gym retro," GitHub repository, 2019, <https://github.com/openai/retro>.
- [7] K. Cobbe *et al.*, "Phasic policy gradient," *ICML*, 2020.
- [8] A. Raffin *et al.*, "Stable-baselines3," 2021, [GitHub repository](https://github.com/DLR-RM/stable-baselines3).
- [9] P. Dhariwal *et al.*, "Openai baselines," GitHub repository, 2017, <https://github.com/openai/baselines>.
- [10] A. Y. Ng *et al.*, "Policy invariance under reward transformations," in *ICML*, 1999.